Jaberi-Elahi Syndrome (JABELS): A Comprehensive Disease Characterization

Disease: Jaberi-Elahi Syndrome (JABELS) **Causal gene: GTPBP2** (GTP-binding protein 2; HGNC, chromosome 6p21.1; UniProt Q9BX10) **Identifiers:** OMIM #617988 · MONDO:0060711 · Category: Mendelian, autosomal recessive

Summary

Jaberi-Elahi syndrome is an ultra-rare autosomal-recessive neurodevelopmental and neurodegenerative disorder caused by biallelic loss-of-function variants in the gene GTPBP2. GTPBP2 encodes a 602-amino-acid translational (tr-type) GTPase that partners with PELO (Pelota) to rescue stalled ribosomes as part of the ribosome-associated quality-control / no-go decay machinery. The disorder was first delineated in 2016 in a consanguineous Iranian family (Jaberi, Elahi, and colleagues) in which three affected siblings presented with neurodegeneration accompanied by brain iron accumulation, and it has since been confirmed and broadened by multiple independent families worldwide. Fewer than ~30 patients had been reported in the literature as of 2024, making this one of the rarest Mendelian neurodevelopmental disorders characterized to date.

The clinical picture is dominated by early motor developmental delay (≈93% of patients), sparse hair (≈93%), speech disorder (≈86%), intellectual disability (≈82%), hypotonia (≈82%), skeletal anomalies (≈82%), and visual/retinal impairment (≈64%), with additional features including microcephaly (which may be severe and congenital), progressive spasticity, choreoathetoid movements, refractory epilepsy, facial dysmorphism, cerebral/cerebellar atrophy, and—in a subset of families—agenesis of the corpus callosum and brain iron accumulation. Expressivity is notably variable and some features (ectodermal/iron abnormalities) are non-obligate and appear family-specific.

The mechanistic basis is well-supported by mouse and invertebrate models. GTPBP2 loss causes codon-specific ribosome stalling (demonstrated at AGA/Arg codons when a CNS-specific tRNA is also limiting), which triggers the GCN2–eIF2α–ATF4 integrated stress response (ISR). Paradoxically, this ISR is neuroprotective and *attenuates* rather than drives the

neurodegeneration, implying that unresolved stalling and downstream proteostatic collapse are the primary injurious events. The closely related paralog GTPBP1 produces an essentially identical syndrome and acts non-redundantly in the same pathway. There is no disease-specific therapy; management is entirely supportive, and diagnosis rests on exome/genome sequencing demonstrating biallelic GTPBP2 variants.

Key Findings

Finding 1 — JABELS is an autosomal-recessive neurodevelopmental/ neurodegenerative disorder caused by biallelic GTPBP2 loss-of-function (F001)

The disorder was first reported by Jaberi et al. (2016) in a consanguineous Iranian family with three affected siblings. Using linkage analysis plus exome sequencing, they identified a homozygous splice-site mutation in **GTPBP2** that deletes exon 9 and produces a truncated protein lacking conserved C-terminal domains. The authors state: *"A splice site mutation in GTPBP2 that encodes GTP-binding protein 2 was identified in the patients and considered possible cause of their disease"* and note that *"Iron accumulation in the brain of the patients was a notable phenotypic feature"* (PMID: 26675814).

Independent confirmation came from Bertoli-Avella et al. (2018), who reported three unrelated families with homozygous nonsense variants—**c.430C>T p.(Arg144*)**, **c.1219C>T p.(Gln407*)**, and **c.1408C>T p.(Arg470*)**—establishing pathogenicity beyond the index family: *"whole-exome sequencing identified the homozygous non-sense variants c.430[C>T];[C>T] p.(Arg144*), c.1219[C>T];[C>T] p.(Gln407*) and c.1408[C>T];[C>T] p.(Arg470*) in GTPBP2"* (PMID: 29449720). Together these establish a recessive loss-of-function mechanism. **Identifiers:** OMIM #617988; MONDO:0060711; gene GTPBP2 (6p21.1); synonym JABELS.

Finding 2 — Core phenotype spectrum and frequencies (28-patient review) (F002)

The most comprehensive phenotype tabulation comes from Manoochehri et al. (2024), who reviewed 28 reported individuals. The dominant features and their frequencies are summarized below:

Phenotype	Frequency	Suggested HPO term
Motor developmental delay	26/28 (93%)	HP:0001270 Motor delay
Sparse hair	26/28 (93%)	HP:0008070 Sparse hair
Speech disorder	24/28 (86%)	HP:0000750 Delayed speech / HP:0002167 Speech articulation difficulties
Intellectual disability	23/28 (82%)	HP:0001249 Intellectual disability
Hypotonia	23/28 (82%)	HP:0001252 Hypotonia
Skeletal problems	23/28 (82%)	HP:0000924 Abnormality of the skeletal system
Visual impairment	18/28 (64%)	HP:0000505 Visual impairment

Additional reported features include abnormal reflexes, facial dysmorphism, poor growth, microcephaly (HP:0000252), progressive spasticity (HP:0002061), choreoathetoid movements (HP:0001266), refractory epilepsy (HP:0001250 Seizure), brain atrophy (HP:0012444), agenesis of the corpus callosum (HP:0001274), and variable brain iron accumulation (HP:0011392 Abnormal cerebral iron deposition). The authors summarize: *"A large numbers of patients manifest motor development delay (26/28), sparse hair (26/28), and speech disorder (24/28). Moreover, a significant fraction of patients suffer from intellectual disability (23/28), hypotonia (23/28), skeletal problems (23/28), and visual impairment (18/28)"* (PMID: 38852771).

Finding 3 — Mechanism: GTPBP2 is a ribosome-rescue factor; its loss causes codon-specific ribosome stalling and neurodegeneration (F003)

The pivotal mechanistic work came from Ishimura et al. (2014), who showed in mice that **loss of GTPBP2—a binding partner of the ribosome-recycling protein Pelota—combined with mutation of a CNS-specific tRNA gene (n-Tr20) causes ribosome stalling and widespread neurodegeneration**: *"loss of GTPBP2, a novel binding partner of the ribosome recycling protein Pelota, in mice with a mutation in a tRNA gene that is specifically expressed in the central nervous system causes ribosome stalling and widespread neurodegeneration"* (PMID: 25061210).

This paradigm was reinforced by Terrey et al. (2020) studying the paralog GTPBP1: *"Loss of a tRNA gene leads to ribosomal pausing that is resolved by the translational GTPase GTPBP2, and in its absence causes neuron death"* (PMID: 33186095), indicating that GTPBP1 and GTPBP2 act non-redundantly in the same ribosome-rescue pathway. Salpietro et al. (2024) added that human disease variants are loss-of-function with ribosome-profiling abnormalities, and that *Drosophila* knockdown of the ortholog CG2017 recapitulates the defects (PMID: 38118446).

GO terms: GO:0072344 rescue of stalled ribosome; GO:0006412 translation; GO:0003924 GTPase activity. **CL terms:** CL:0000540 neuron; CL:0000210 photoreceptor cell (retinal degeneration).

Finding 4 — GTPBP1 and GTPBP2 produce an identical syndrome; both are translational GTPases related to eEF1A/eRF3/Hbs1 (F004)

Salpietro et al. (2024) described 20 individuals from 16 families with biallelic GTPBP1 **or** GTPBP2 variants sharing *"microcephaly, profound neurodevelopmental impairment, pathognomonic craniofacial features, and ectodermal defects"*, together with abnormal vision/hearing, progressive spasticity, choreoathetoid movements, refractory epilepsy, and brain atrophy; all variants were loss-of-function (PMID: 38118446).

Zinoviev et al. (2018) classified these proteins biochemically: *"GTP-binding protein 1 (GTPBP1) and GTPBP2 comprise a divergent group of translational GTPases with obscure functions, which are most closely related to eEF1A, eRF3, and Hbs1"* (PMID: 30108131). GTPBP1 has eEF1A-like aminoacyl-tRNA delivery activity and stimulates exosomal mRNA degradation, implicating both paralogs in mRNA surveillance and ribosome-associated quality control (RQC).

Finding 5 — Downstream mechanism: AGA-codon stalling activates the neuroprotective GCN2–eIF2α–ATF4 integrated stress response (F005)

Ishimura et al. (2016) dissected the downstream signaling in *Gtpbp2(nmf205)* mice: *"neuronal elongation complexes are stalled at AGA codons due to deficiencies in a tRNA(Arg)UCU tRNA and GTPBP2, a mammalian ribosome rescue factor. Increased levels of phosphorylation of eIF2α (Ser51) were detected prior to neurodegeneration"* (PMID: 27085088). Transcriptomics showed ATF4 activation—the hallmark of the integrated stress response (ISR). Critically, the authors found that *"this pathway was activated by the eIF2α kinase, GCN2, in an apparent deacylated tRNA-independent fashion. Further we found that the ISR attenuates neurodegeneration"* (PMID: 27085088).

This is a mechanistically important nuance: the ISR is a **protective, compensatory** response, not the proximate cause of neuron death. Neurodegeneration therefore reflects the failure/insufficiency of ribosome rescue and proteostasis, with the ISR buffering—but not preventing—the injury. **GO terms:** GO:0140469 GCN2-mediated ISR signaling; GO:0034198 cellular response to amino acid starvation.

Finding 6 — Model organisms and epidemiology (F006)

Mouse: The *Gtpbp2(nmf205)* mouse on the B6J background (which carries the n-Tr20 tRNA^Arg mutation) develops progressive neurodegeneration, retinal degeneration, and ribosome stalling (PMID: 25061210; PMID: 27085088). **Invertebrate:** *Drosophila* knockdown of the ortholog **CG2017** recapitulates defects: *"we also investigated the effect of reducing expression of CG2017, an ortholog of human GTPBP1/2, in the fruit fly Drosophila melanogaster"* (PMID: 38118446).

Epidemiology: JABELS is ultra-rare, with ~28 individuals reported as of 2024 (PMID: 38852771). GTPBP2 is listed among autosomal-recessive NBIA (neurodegeneration with brain iron accumulation) genes with a very low combined lifetime risk. Kolarova et al. (2022) estimated: *"The combined estimated lifetime risk of all 13 investigated NBIA disorders is 0.88 (95% confidence interval 0.70-1.10) per 100,000 based on the global gnomAD dataset"*, with GTPBP2 individually far below the top contributors (PLA2G6/PANK2/COASY) (PMID: 35180557).

Finding 7 — Expanded phenotypic spectrum: severe congenital microcephaly, early seizures, white-matter and cardiac anomalies (F007)

Soğukpınar et al. (2026) reported a 4-month-old female of consanguineous parents with a novel homozygous frameshift GTPBP2 variant: *"presenting with severe microcephaly, developmental delay, hypotonia, and craniofacial features. Additional findings included left-sided pes equinovarus, secundum atrial septal defect, and periventricular white matter abnormalities on brain magnetic resonance imaging (MRI). No overt ectodermal abnormalities were observed"* (PMID: 42553919). This broadens the spectrum to include **cardiac (secundum ASD) and limb (pes equinovarus) anomalies** and highlights that ectodermal features are non-obligate.

Bertoli-Avella et al. (2018) similarly emphasized family-specific expressivity: *"Notable differences include structural brain abnormalities (e.g., agenesis of the corpus callosum, exclusive to our patients), and evidence for brain iron accumulation (exclusive to the previously described family)"* (PMID: 29449720). **HPO terms:** HP:0001631 Atrial septal defect; HP:0001762 Talipes equinovarus; HP:0002500 Abnormal cerebral white matter morphology; HP:0000252 Microcephaly.

Finding 8 — GTPBP2 protein architecture (F008)

Per UniProt Q9BX10, human GTPBP2 is a **602-amino-acid** protein containing a tr-type (translational) G domain (≈residues 170–398) with GTP-binding/P-loop motifs (binding sites ~179–186, 260–264, 316–319) and an N-terminal disordered region (16–64). Functionally it is involved in the rescue of ribosome stalling due to the presence of non-functional tRNA and has very low GTP-binding activity; it interacts with PELO (Pelota). The weak/context-dependent GTP binding is supported by Zinoviev et al.: *"Although GTPBP2's binding to GTP was stimulated by Phe-tRNA"* (PMID: 30108131). Tissue expression is highest in thymus, spleen, and testis, with lower expression in brain, lung, kidney, and ovary. Disease-causing truncating variants (p.Arg144*, p.Gln407*, p.Arg470*, exon-9 splice deletion) remove or disrupt the G domain and/or C-terminal effector region. **GO cellular component:** GO:0005737 cytoplasm; GO:0022626 cytosolic ribosome (associated).

Finding 9 — gnomAD constraint confirms a recessive loss-of-function gene with strong missense intolerance (F009)

Population constraint metrics for GTPBP2 (ENSG00000172432, chr6:43,605,316–43,629,264) are consistent with a recessive disease gene:

Metric	Value	Interpretation
pLI	0.0002	NOT haploinsufficient — heterozygous LoF tolerated
LOEUF (oe_lof upper)	0.68	Moderate LoF constraint
observed/expected LoF	0.51 (obs 32 / exp 63); LoF Z = 3.31	Depleted for LoF
missense Z	4.61	**Strong** missense constraint
oe_mis	0.62	Missense depleted

The low pLI indicates heterozygous loss-of-function alleles are tolerated in the general population (carriers are unaffected), consistent with recessive inheritance, whereas the high missense Z shows the coding sequence—especially the G domain—is under strong purifying selection.

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic loss-of-function variant in GTPBP2** (nonsense, frameshift, or splice) → **absent or truncated, non-functional GTPBP2 protein** lacking the tr-type G domain and/or C-terminal effector region (*demonstrated*;

   P 26675814

   P 29449720

).

   P 42553919

2. Loss of functional GTPBP2 → **failure of PELO-dependent rescue of stalled/paused ribosomes** (ribosome-associated quality control / no-go decay) (*demonstrated in mouse*;

).

   P 25061210

3. Impaired rescue, when combined with limiting tissue-specific tRNA supply (e.g., CNS-enriched tRNA^Arg_UCU), → **codon-specific ribosome stalling at cognate codons (AGA/Arg)** and accumulation of stalled elongation complexes (*demonstrated in mouse*;

).

   P 27085088

4. Ribosome stalling → **GCN2 (eIF2α kinase) activation → eIF2α-Ser51 phosphorylation → ATF4-driven integrated stress response (ISR)** (*demonstrated*;

).

   P 27085088

   Branch: the ISR is **neuroprotective** and *attenuates* degeneration, so it is a compensatory side-arm, not the injurious main path.

5. Persistent unresolved stalling, aberrant/truncated nascent peptides, and proteostatic stress (*inferred* for human disease from model data) → **progressive neuronal and retinal cell death** (*demonstrated in mouse*;

   P 25061210

).

   P 27085088

6. Neurodegeneration + abnormal neurodevelopment → **clinical phenotype**: motor delay, intellectual disability, speech disorder, hypotonia, spasticity, choreoathetosis, epilepsy,

visual impairment, microcephaly, brain atrophy (*demonstrated in humans*;

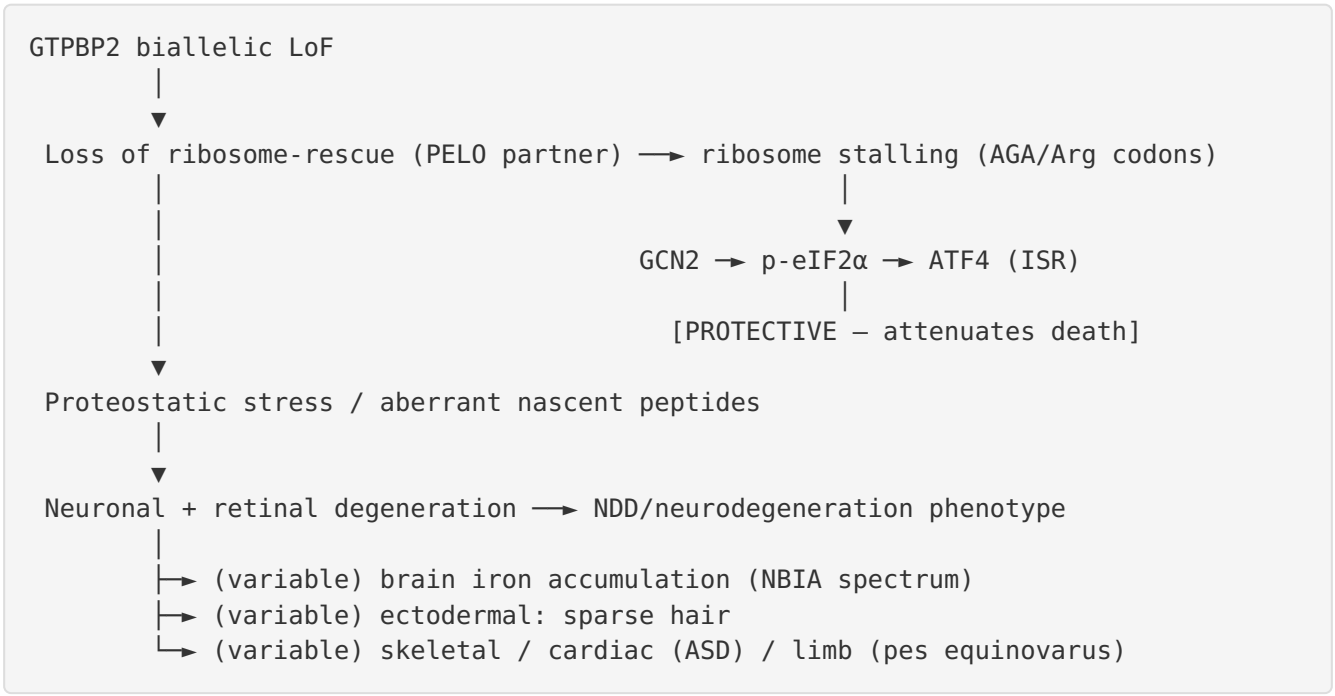
P 38852771

P 38118446

P 29449720

).

7. **Parallel/variable branches** (family-specific, mechanism *inferred*): brain iron accumulation (NBIA-spectrum), ectodermal defects (sparse hair), skeletal/cardiac/limb anomalies—reflecting broader roles of translational quality control in development.



Upstream vs downstream: The mutation and loss of ribosome rescue are upstream; stalling is the central lesion; the GCN2/ISR arm and neuronal death are downstream. The ISR is best understood as a downstream *buffer*. **Cell types:** neurons (CL:0000540), retinal photoreceptors (CL:0000210), cerebellar/cortical neurons. **Processes:** GO:0072344 (rescue of stalled ribosome), GO:0006412 (translation), GO:0140469 (GCN2-mediated ISR).

Section-by-Section Knowledge Base Content

1. Disease Information

JABELS is an autosomal-recessive neurodevelopmental/neurodegenerative syndrome. **Identifiers:** OMIM #617988; MONDO:0060711; gene GTPBP2 (HGNC). A robust ICD-10/ICD-11, MeSH, and dedicated Orphanet code are not well established for this ultra-rare entity (information not well-indexed). **Synonyms:** JABELS; Jaberi-Elahi syndrome; GTPBP2-related neurodegeneration with brain iron accumulation (NBIA-spectrum). Information is derived from aggregated, disease-level case reports and small family series (not EHR-scale cohorts).

2. Etiology

Causal factor: purely genetic—biallelic (homozygous or compound-heterozygous) loss-of-function variants in GTPBP2. **Genetic risk factor:** consanguinity (most reported families are consanguineous, enriching for homozygous recessive alleles). **Environmental/protective factors and gene–environment interactions:** none established (not applicable for a monogenic recessive disorder). No protective variants are known.

3. Phenotypes

See Finding 2 table for core phenotypes and frequencies. Phenotype **types** span clinical signs (hypotonia, spasticity), physical manifestations (sparse hair, microcephaly, dysmorphism, skeletal/cardiac/limb anomalies), behavioral/cognitive changes (intellectual disability, speech disorder), and neuroimaging/laboratory abnormalities (brain atrophy, white-matter changes, brain iron accumulation). **Onset:** neonatal to infantile; developmental delay is typically apparent in the first months of life. **Severity:** moderate to severe. **Progression:** progressive/neurodegenerative with spasticity and sometimes refractory epilepsy. **QoL impact:** profound—most affected individuals have severe intellectual and motor disability requiring full-time care.

4. Genetic / Molecular Information

Causal gene: GTPBP2 (6p21.1). **Reported pathogenic variants:** c.430C>T p.(Arg144*); c.1219C>T p.(Gln407*); c.1408C>T p.(Arg470*) (all nonsense,);

P 29449720

exon-9 splice-site deletion

( ); P 26675814

novel homozygous frameshift

( ). P 42553919

Variant types: nonsense, frameshift, splice-site—all truncating/LoF. **Classification:** pathogenic/likely pathogenic per ACMG (PVS1 loss-of-function + recessive segregation). **Allele frequency:** individually absent/ultra-rare in gnomAD. **Origin:** germline. **Functional consequence:** loss of function. **Modifier genes:** tissue-specific tRNA abundance (e.g., n-Tr20 tRNA^Arg in mouse) is a demonstrated genetic modifier of ribosome stalling; GTPBP1 is a non-redundant paralog. **Epigenetics/chromosomal abnormalities:** none reported.

5. Environmental Information

No environmental, lifestyle, or infectious contributors. Not applicable.

6. Mechanism / Pathophysiology

See the ordered causal chain and diagram above (Findings 3–5, 8).

7. Anatomical Structures Affected

Primary: central nervous system—cerebral cortex (UBERON:0000956), cerebellum (UBERON:0002037), basal ganglia (iron deposition; UBERON:0002420), white matter (UBERON:0002316), corpus callosum (UBERON:0002336, agenesis in some). **Secondary/other systems:** retina/eye (UBERON:0000970), integument/hair follicle (ectodermal: sparse hair; UBERON:0002073), skeletal system (UBERON:0001434), heart (atrial septum; UBERON:0002085). **Cells:** neurons (CL:0000540), photoreceptors (CL:0000210). **Subcellular:** cytoplasmic translation machinery / ribosome (GO:0022626); abnormal iron handling implicates mitochondria/lysosomes (inferred). **Lateralization:** generally bilateral/symmetric CNS involvement; one patient had left-sided pes equinovarus (asymmetric limb finding).

8. Temporal Development

Onset: congenital/infantile (developmental delay in first months). **Course:** chronic, progressive/neurodegenerative, lifelong. **Stages:** early developmental delay → progressive spasticity, movement disorder, epilepsy → advanced neurodegeneration with brain atrophy. No remission. Critical period for any future therapy would plausibly be early, before irreversible neuronal loss (inferred).

9. Inheritance and Population

Inheritance: autosomal recessive. **Penetrance:** appears complete in biallelic LoF carriers. **Expressivity:** variable—iron accumulation, corpus callosum agenesis, ectodermal, and cardiac/limb features are family-specific. **Consanguinity:** strong role; most families are consanguineous. **Carrier frequency:** very low (gnomAD; low pLI = heterozygotes tolerated). **Epidemiology:** ultra-rare, ~28 patients reported; part of the recessive NBIA group whose combined lifetime risk is ~0.88/100,000 ( ). P 35180557

Geography: reported in Iranian, Middle Eastern/North African, and other consanguineous populations; no clear sex predilection.

10. Diagnostics

Genetic testing is the diagnostic cornerstone: whole-exome or whole-genome sequencing (often with homozygosity mapping in consanguineous pedigrees) identifying biallelic GTPBP2 LoF variants; single-gene/panel testing (NBIA or intellectual-disability panels including GTPBP2) is an alternative. **Imaging:** brain MRI shows atrophy, white-matter abnormalities, possible corpus callosum agenesis, and T2*/SWI signal consistent with iron accumulation in a subset. **Supportive studies:** EEG for epilepsy; ophthalmologic/retinal evaluation; echocardiography if cardiac anomaly suspected. **No specific biochemical biomarker** exists. **Differential diagnosis:** other NBIA disorders (PKAN/PANK2, PLAN/PLA2G6, MPAN/C19orf12, BPAN/WDR45), GTPBP1-related disorder (phenotypically identical), and other recessive neurodevelopmental/microcephaly syndromes.

11. Outcome / Prognosis

Prognosis is poor: severe, lifelong intellectual and motor disability with progressive neurologic decline. Formal survival/mortality statistics are unavailable given the small cohort, but refractory epilepsy and severe neurodisability imply increased morbidity and likely reduced life expectancy. No recovery potential; management is palliative/supportive.

12. Treatment

No disease-specific or disease-modifying therapy exists. Management is **supportive and multidisciplinary:** anti-seizure medications for epilepsy; physical/occupational/speech therapy; management of spasticity; nutritional support; ophthalmologic and orthopedic/cardiac care as needed. The protective role of the GCN2/ISR arm

(P 27085088) suggests ISR modulation as a *theoretical* future target, and tRNA-supplementation strategies (cf. tRNA overexpression rescuing related tRNA/translation disorders, P 34516840) are conceptual avenues—none clinically available. **NCIT terms:** NCIT:C15313 Supportive Care; NCIT:C15329 Rehabilitation Therapy; NCIT:C98015 Antiepileptic Agent.

13. Prevention

Primary prevention is limited to **genetic counseling** for consanguineous/at-risk families, carrier testing, and reproductive options including prenatal diagnosis and preimplantation genetic testing once the familial variant is known. No vaccines, behavioral, or public-health interventions apply.

14. Other Species / Natural Disease

No naturally occurring companion-animal or wildlife disease is documented. **Orthologs:** mouse *Gtpbp2*, *Drosophila* CG2017. The mechanism (ribosome rescue/RQC) is deeply evolutionarily conserved. No zoonotic potential.

15. Model Organisms

- **Mouse:** *Gtpbp2(nmf205)* loss-of-function on the B6J background (which harbors the modifier n-Tr20 tRNA^Arg mutation) models progressive neurodegeneration, retinal degeneration, AGA-codon ribosome stalling, and GCN2/ISR activation (P 25061210). P 27085088 **Recapitulation:** strong for neurodegeneration, retinal involvement, and the core molecular mechanism; the requirement for a co-occurring tRNA modifier is a caveat for direct human translation.
- **Invertebrate:** *Drosophila* CG2017 knockdown recapitulates GTPBP1/2 defects (P 38118446).
- **Databases:** MGI (mouse Gtpbp2), FlyBase (CG2017), Alliance of Genome Resources.

Evidence Base

PMID	Title (abbrev.)	Evidence type	Role in this report
26675814	Mutation in GTPBP2 in family w/ neurodegeneration + brain iron	Human clinical	Original description; causal gene; iron phenotype (F001)
29449720	Biallelic inactivating GTPBP2 variants cause NDD	Human clinical	Independent confirmation; 3 nonsense variants; variable expressivity (F001, F007)
38852771	Novel GTPBP2 mutation + literature review	Human clinical review	28-patient phenotype frequencies; epidemiology (F002, F006)
25061210	Ribosome stalling from tRNA mutation causes neurodegeneration	Mouse model	Core mechanism: GTPBP2 = ribosome rescue factor (F003)
27085088	GCN2 activation by ribosome stalling	Mouse model	AGA stalling; eIF2α/ATF4 ISR; protective (F005)
33186095	GTPBP1 resolves paused ribosomes	Mouse model	Paralog; non-redundant pathway (F003, F004)
38118446	GTPBP1/GTPBP2 cause identical NDD syndrome	Human + Drosophila	Syndrome definition; LoF; fly model (F004, F006)
30108131	Functions of GTPBP1/GTPBP2	In vitro/ biochemical	Protein family (eEF1A/eRF3/ Hbs1); weak GTP binding (F004, F008)
35180557	Lifetime risk of recessive NBIA disorders	Computational/ genetic	Epidemiology context (F006)
42553919	Novel homozygous frameshift GTPBP2	Human clinical	Expanded phenotype: cardiac/ limb; no ectodermal (F007)
39419454	Metabolic alterations in MPAN/NBIA fibroblasts	In vitro	Places GTPBP2 in NBIA iron/ lipid-metabolism context
34516840	tRNA overexpression rescues neuropathy	Mouse model	Conceptual therapeutic analogy (tRNA supplementation)
29096076		Review	

PMID	Title (abbrev.)	Evidence type	Role in this report
	Regulation of mRNA translation in neurons		Broader context: translation dysregulation in neurodegeneration
25061188	Ribosome rescue and neurodegeneration	Commentary	Contextual commentary on P 25061210

Evidence strength is high for the causal gene, LoF mechanism, core phenotype, and ribosome-rescue pathway (concordant human genetics + mouse + invertebrate data). The GCN2/ISR arm is directly demonstrated in mouse; its exact role in human patients is inferred.

Limitations and Knowledge Gaps

- **Very small human cohort (~28 patients).** Frequency estimates, natural-history staging, survival, and genotype–phenotype correlations are statistically fragile.
- **Model caveat:** the pivotal mouse neurodegeneration phenotype required a *co-occurring* CNS-tRNA modifier (n-Tr20). Whether analogous tissue-specific tRNA limitations drive the human phenotype is unproven; it is an attractive but inferred bridge.
- **Mechanism of secondary features** (brain iron accumulation, ectodermal/sparse hair, skeletal/cardiac/limb anomalies) is not mechanistically explained and is family-variable.
- **No validated biomarker** for diagnosis, severity, or progression monitoring.
- **No human therapeutic data**; ISR-modulating or tRNA-based strategies are purely conceptual.
- **Ontology/registry indexing** (ICD-11, MeSH, Orphanet code) for JABELS is incomplete.

Proposed Follow-up Experiments / Actions

1. **Establish an international patient registry** to pool genotype–phenotype data, natural history, and survival from the dispersed case reports.

2. **Patient-derived models:** generate iPSC-derived neurons and cerebral organoids from GTPBP2-null patients to test whether codon-specific stalling and ISR activation occur in human cells without an engineered tRNA modifier.
3. **Ribosome profiling in human GTPBP2-deficient neurons** to map stall sites and compare with the mouse AGA-codon signature.
4. **Test ISR modulators** (e.g., ISRIB, GCN2 agonists/antagonists) in patient neurons/organoids and the *Gtpbp2* mouse to determine whether enhancing the protective ISR or clearing stalled ribosomes is beneficial.
5. **tRNA-supplementation / AAV gene-replacement proof-of-concept** in the *Gtpbp2* mouse, leveraging the analogy to tRNA rescue of tRNA-synthetase neuropathy (P 34516840).
6. **Mechanistic study of brain iron accumulation** in GTPBP2 deficiency to clarify its link (if any) to impaired translational quality control and to position JABELS within the NBIA spectrum.
7. **Formal ClinVar/ClinGen curation** and ACMG classification of all reported GTPBP2 variants, plus submission of an Orphanet/ICD-11 entry to improve indexing.

Report compiled from 5 discovery iterations, 9 confirmed findings, and 14 reviewed papers.
